

Lecture 3: Point Processing - Part 1

In Image Processing, techniques are mainly divided into two types:

- Spatial Domain Image Processing
- Frequency Domain Image Processing

Typical image processing pipeline



1. Spatial Domain Image - $f(x, y)$

- The first image is in the **spatial domain**.
- It is represented by pixel coordinates (x, y) .
- Each point contains an intensity/brightness value.

2. Transform $\rightarrow$ Frequency Domain - $F(u, v)$

- The image is transformed using a mathematical transform like the Fourier Transform.
- Now it is represented in the **frequency domain** as $F(u, v)$.
- (u, v) represent frequency components instead of spatial position.

3. Processing in Frequency Domain

- Modify frequency components:
 - Remove noise (low-pass filter)
 - Enhance edges (high-pass filter)
 - Image compression
- Processing is often easier in the frequency domain for filtering.

4. Inverse Transform → Back to Spatial Domain

- After processing, apply the inverse Fourier transform.
- Convert back to spatial image $g(x, y)$.
- This is the final processed image.

Image $f(x,y) \rightarrow$ Transform $\rightarrow F(u,v) \rightarrow$ Frequency processing $\rightarrow$ Inverse transform $\rightarrow g(x,y)$.

If we directly process in the spatial domain, then we actually do not need any transform.

In that case, we use what is called the Identity (Unity) Transform.

What is Unity (Identity) Transform?

It means: $T[f(x,y)] = g(x,y)$

Here, $g(x,y)=f(x,y)$ $[T=I]$

So the image remains in the **same spatial domain** (x, y) .

There is **no change of domain**.

In Spatial Domain Processing:

- Input image $\rightarrow f(x,y)$
- Output image $\rightarrow g(x,y)$
- $T[] =$ Mapping Function

Example: $g(x,y) = f(x,y) + c$

(adding brightness)

No Fourier transform is needed.

So Pipeline Becomes:

Spatial Image $\rightarrow$ Processing $\rightarrow$ Output Image

(Transform step = Identity / Unity transform)

Scope of Image Processing

1. Image Enhancement

- Improve the **visual quality** of an image.
- Make the image clearer, sharper, brighter, or remove noise.
- Used to make the image better for human viewing.

Examples:

Contrast stretching, smoothing, sharpening, histogram equalization.

2. Improved Recognition Rate

- Improve the performance of **machine recognition systems**.
- Helps in better detection, classification, and identification.
- Used in face recognition, medical diagnosis, object detection, etc.

Types of Spatial Image Processing

1. Point Processing

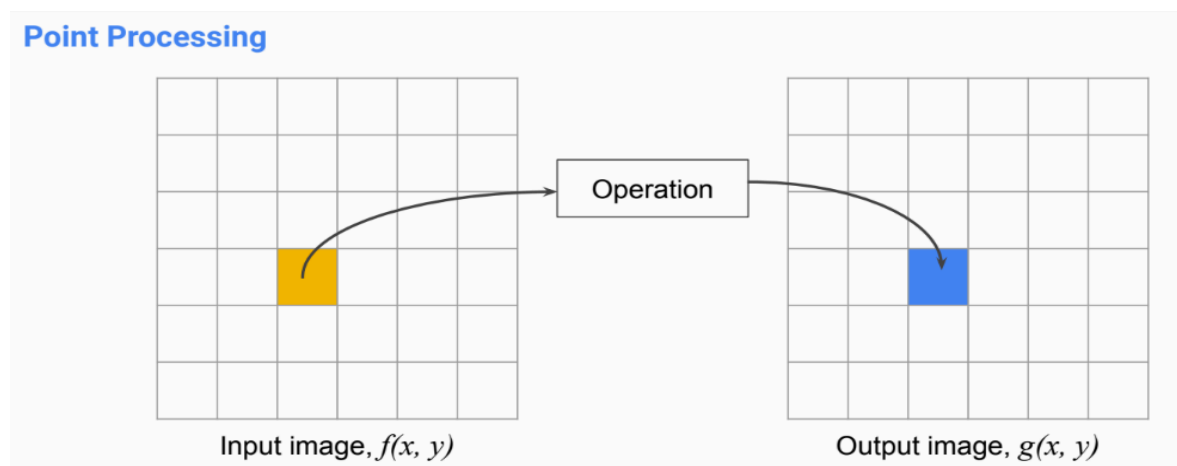
- Operation is applied **individually to each pixel**.

- The output pixel depends only on the **corresponding input pixel**.

$$g(x,y) = T[f(x,y)]$$

Examples:

- Brightness adjustment
- Contrast stretching
- Image negative

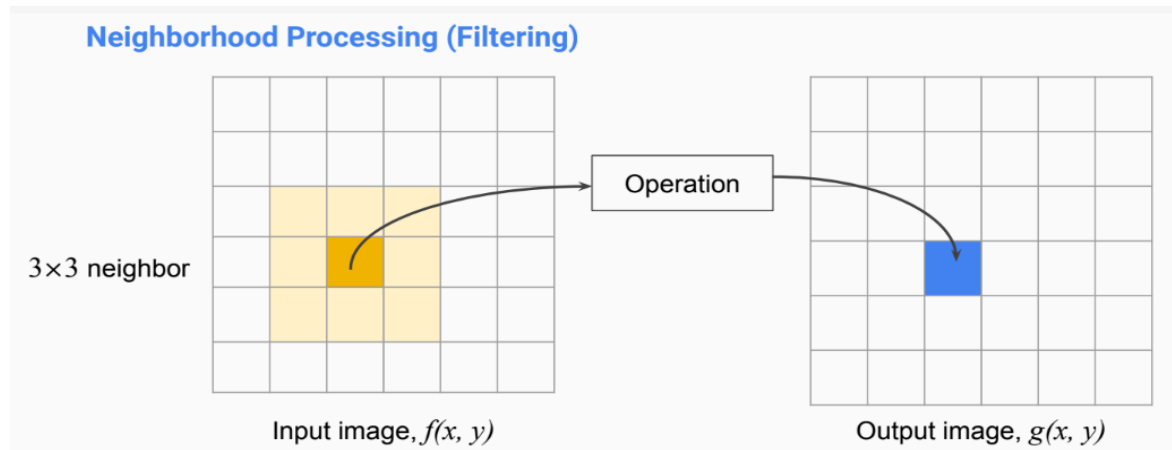


2. Neighbourhood Processing (Local Processing)

- Output pixel depends on the **surrounding pixels (neighbors)**.
- Uses a small window (e.g., 3×3 , 5×5 mask).

Examples:

- Smoothing (mean filter)
- Sharpening
- Edge detection

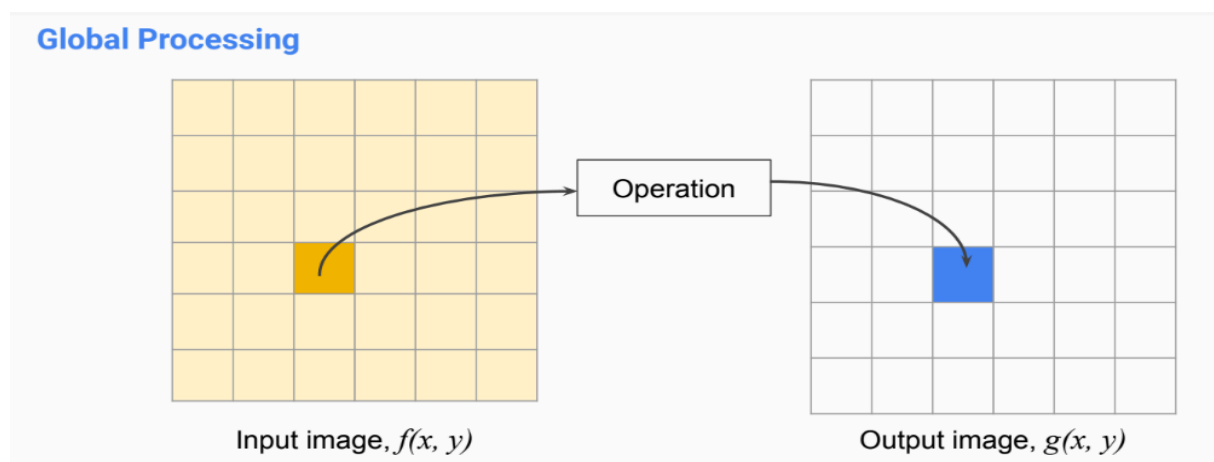


3. Global Processing

- Output depends on the **entire image**.
- Uses overall image information (global statistics).

Examples:

- Histogram equalization
- Global thresholding



Intensity Transformation (Point Processing)

In spatial image processing, intensity transformation is written as:

$$s = T[r]$$

where:

- r = input intensity (input pixel value)
- s = output intensity (processed pixel value)
- $T[]$ = mapping (transformation) function

The function T maps input gray level $r \rightarrow s$.

Graphical Representation

Mapping functions can be visualized using a graph:

- x-axis $\rightarrow$ input intensity (r)
- y-axis $\rightarrow$ output intensity (s)

So each input value is transformed into a new output value.

Inverse Mapping

$$s = 255 - r$$

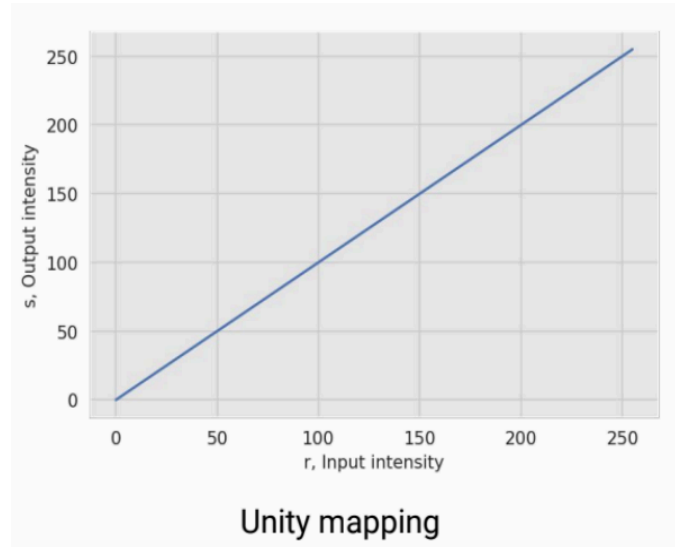
Examples:

- $0 \rightarrow 255$
- $1 \rightarrow 254$
- $255 \rightarrow 0$

Unity (Identity) Mapping

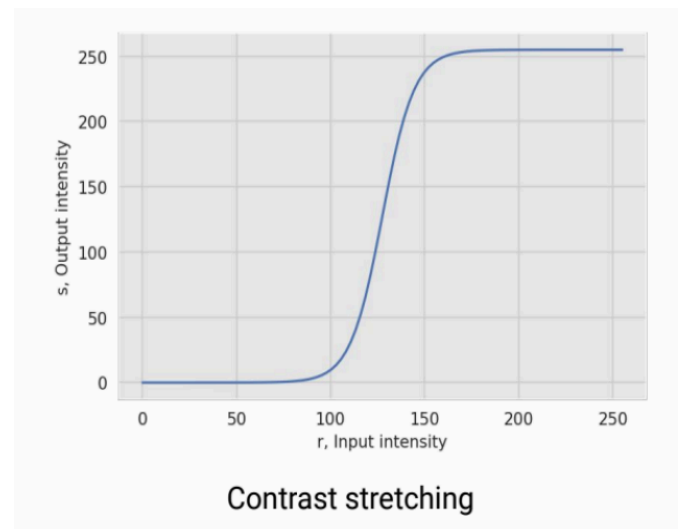
$$s = r$$

- Output image remains the same as input.
- No change in intensity.



Contrast Stretching

- Expands intensity range.
- Improves image contrast.
- Makes dark regions darker and bright regions brighter.



Normalized Intensity Range

For simplicity, intensity values are often normalized:

$$r' = r / 255$$

So intensity range becomes: $[0, 255] \rightarrow [0, 1]$

This makes mathematical analysis easier.

Arithmetic Operations (Brightness Adjustment)

Brightness of an image can be adjusted using **addition and subtraction** operations in spatial domain processing.

Intensity Transformation

$$s = r \pm C$$

where:

- **r** = input intensity
- **s** = output intensity
- **C** = positive constant
- **Addition (+C)** $\rightarrow$ increases brightness
- **Subtraction (-C)** $\rightarrow$ decreases brightness

Intensity values are normalized within the range **[0, 1]**.

Clipping

After addition or subtraction, pixel values may go outside the valid intensity range **[0, 1]**. Since intensity values cannot exceed the allowed limits, **clipping** is applied.

- Maximum intensity = **1** (or 255 in 8-bit image)
- Minimum intensity = **0**

If:

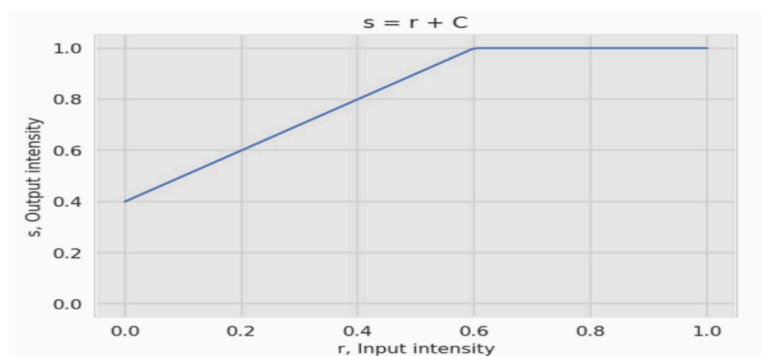
- $(s > 1) \rightarrow \text{set } (s = 1)$
- $(s < 0) \rightarrow \text{set } (s = 0)$

Brightness Adjustment Example ($C = 0.4$)

1. Addition Case: $(s = r + 0.4) \rightarrow$ Brightness increases.

Input (r)	Calculation	Output (s)	After Clipping
0.2	$0.2 + 0.4$	0.6	0.6
0.5	$0.5 + 0.4$	0.9	0.9
0.8	$0.8 + 0.4$	1.2	1 (clipped)

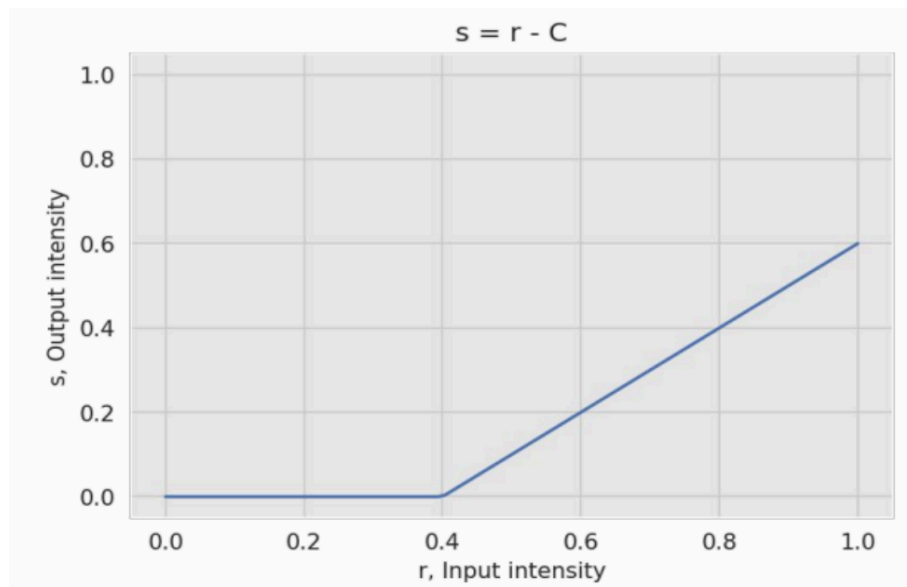
If the value exceeds 1, it is clipped to 1.



2. Subtraction Case: ($s = r - 0.4$) → Brightness decreases.

Input (r)	Calculation	Output (s)	After Clipping
0.7	$0.7 - 0.4$	0.3	0.3
0.4	$0.4 - 0.4$	0	0
0.2	$0.2 - 0.4$	-0.2	0 (clipped)

If the value becomes negative, it is clipped to 0.



Saturation in Brightness Adjustment

During arithmetic brightness operations:

1. Addition

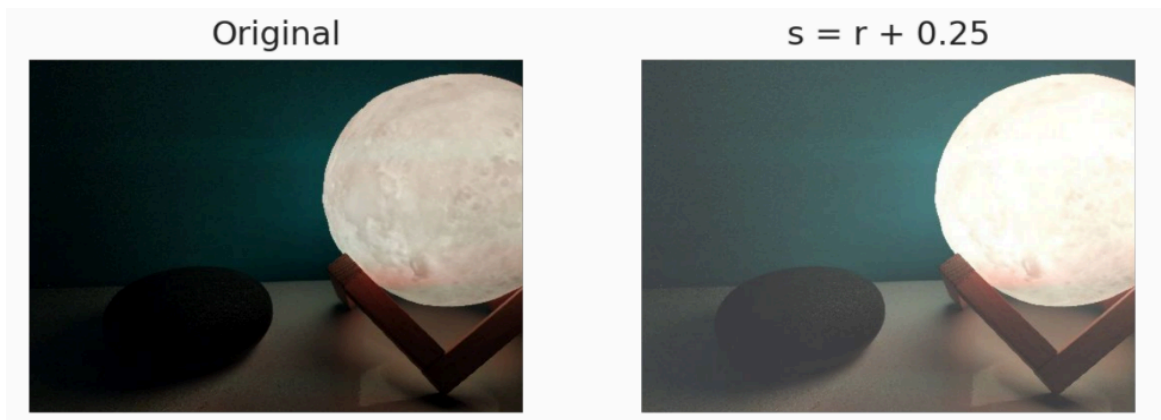
- **Addition increases intensity values.**
- High intensity pixels may exceed the maximum allowed value.
- These values are clipped to the maximum intensity.

Addition might saturate high intensity levels

(bright regions become fully white).

Example:

$$s = r + C, r = 0.9, C = 0.3 \Rightarrow s = 1.2 \rightarrow 1$$



2. Subtraction

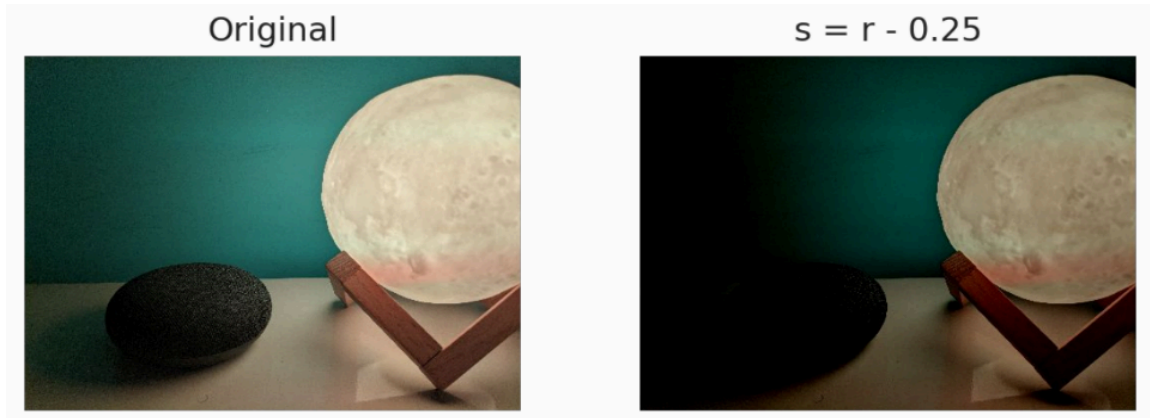
- **Subtraction decreases intensity values.**
- Low intensity pixels may go below zero.
- These values are clipped to the minimum intensity.

Subtraction might saturate low intensity levels

(dark regions become fully black).

Example:

$$S = r - C, r = 0.2, C = 0.4 \Rightarrow s = -0.2 \rightarrow 0$$



Arithmetic Operations - Contrast Adjustment

Contrast of an image can be adjusted using **multiplication and division** operations.

Intensity Transformation

$$s = r \times C \text{ or } s = r / C$$

where:

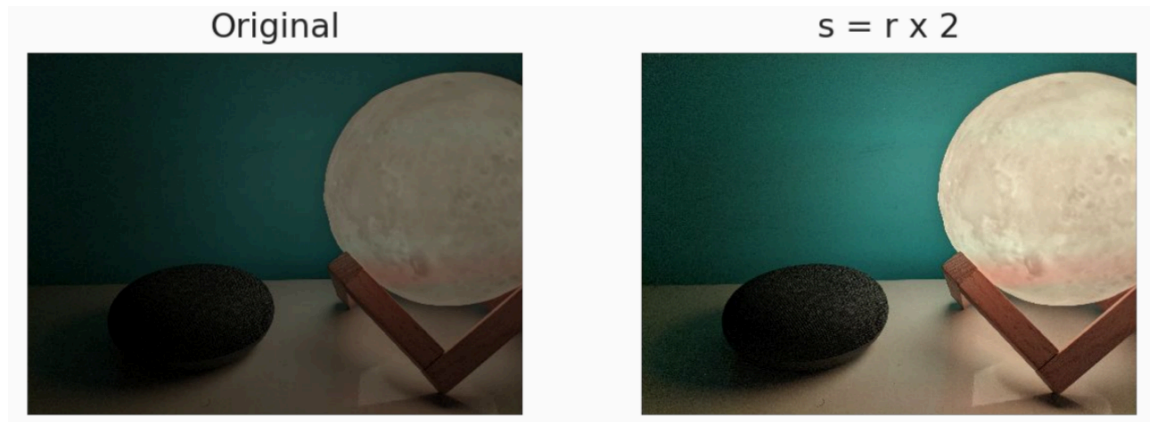
- r = input intensity
- s = output intensity
- C = positive constant

Effect on Image

1. Multiplication ($s = r \times C$)

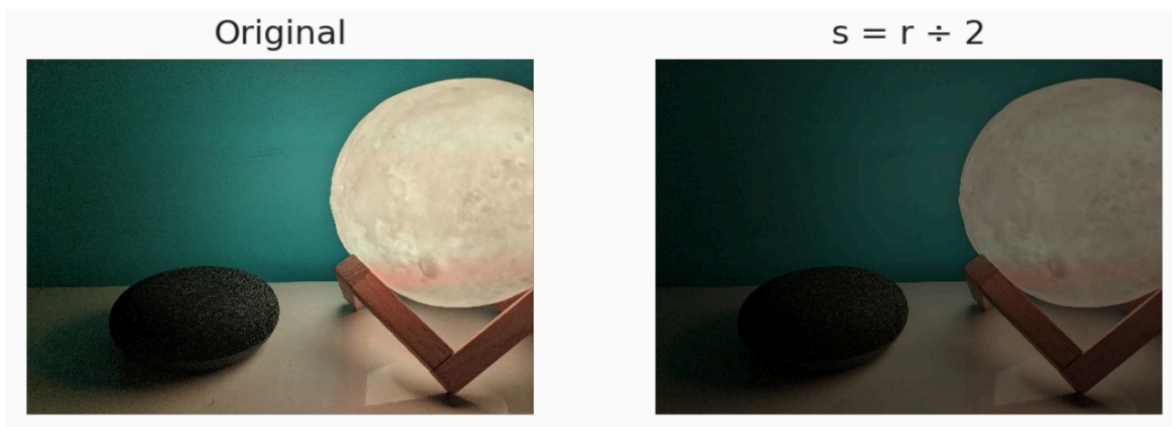
- If $C > 1 \rightarrow$ contrast increases (image becomes more vivid).
- If $0 < C < 1 \rightarrow$ contrast decreases (image becomes dull).

Pixel differences are amplified or reduced proportionally.



2. Division ($s = r / C$)

- Reduces intensity values when ($C > 1$).
- Often used to lower contrast or brightness.



Clipping

Output intensity must remain within the valid range: $[0,1]$

If:

- ($s > 1$) $\rightarrow$ clipped to 1

- $(s < 0) \rightarrow$ clipped to **0**

Clipping ensures valid pixel values.

Contrast Stretching:

Contrast Stretching is an **image enhancement technique** that improves image visibility by expanding the range of pixel intensity values.

Main Idea

- Pixel values are rescaled to use the full available intensity range.
- This increases contrast and **makes details in low-contrast** images more visible.

For example:

- If an image's pixel values lie between **80 and 150**
- And the display range is **0 to 255**
- Contrast stretching maps:
80 $\rightarrow$ 0
150 $\rightarrow$ 255

Key Points

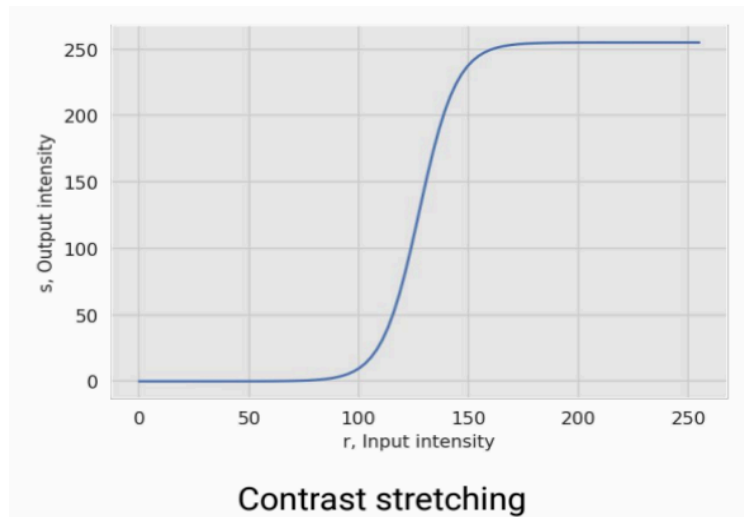
1. Wide range near black/white is compressed

- Very dark and very bright intensity values are squeezed into a smaller range.
- This prevents extreme brightness or darkness from dominating the image.

2. Small range near the middle is expanded

- Mid-level gray values are stretched over a wider range.
- As a result, small intensity differences become more noticeable.

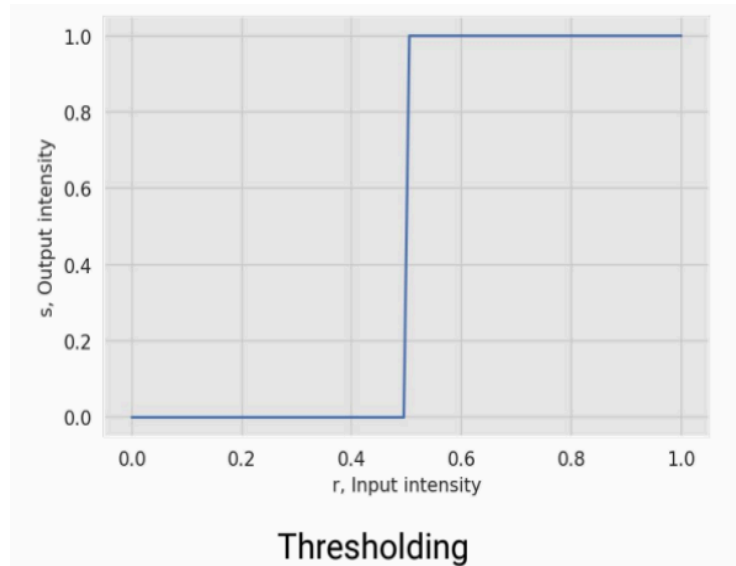
This improves visibility of objects and textures.



Limiting Case: Thresholding

- When contrast stretching becomes extreme, it turns into **thresholding**.
- In thresholding:
 - Pixels above a certain value → **white (1)**
 - Pixels below that value → **black (0)**

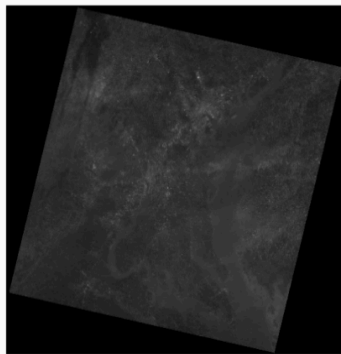
The result is a **binary image** (only black and white).



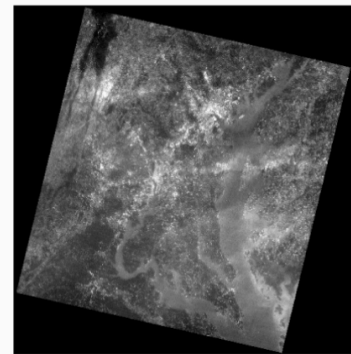
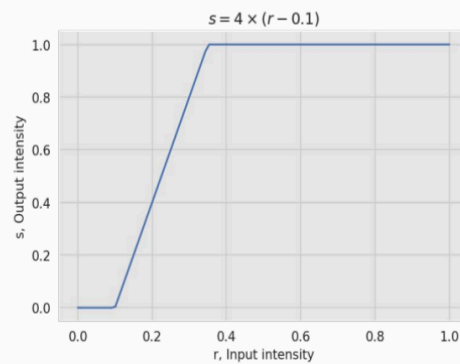
$$s = C1 \times (r - C2)$$

where:

- r = input pixel intensity
- s = output pixel intensity
- $C1, C2$ = constants controlling scaling and shift



Input image (LANDSAT band 1)



Output image

- Most pixel intensities are **very low**, say between **0 and 0.1** (on a 0-1 scale).
- Directly displaying this image would make it **almost completely black**, with very little visible detail.

Step 1: Clip very dark pixels ($0 \rightarrow 0.1$ mapped to 0)

- All pixels from **0 to 0.1** are set to **0** (pure black).
- **Effect:**
 - Extreme dark pixels stay black.
 - Very slight differences in this range are lost.
 - This is the “compression of the low-intensity range” mentioned in contrast stretching.

Step 2: Stretch the mid-low intensities ($0.1 \rightarrow 0.35$)

- Pixels in the range **0.1 to 0.35** are **linearly stretched** to cover a wider output range, e.g., $0 \rightarrow 1$.
- **Effect:**
 - Mid-level dark pixels now occupy the full visible intensity range.
 - The image becomes brighter, and details that were hidden in the dark regions become visible.
 - This is the **main purpose of contrast stretching**: making subtle differences noticeable.

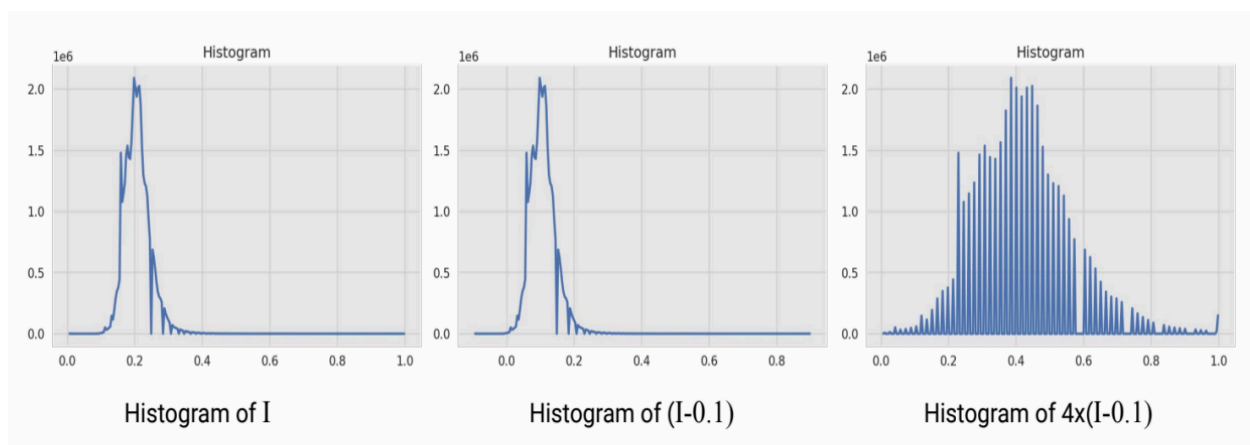
Step 3: Clip high intensities ($0.35 \rightarrow 1$ mapped to 1)

- Any pixels brighter than 0.35 are mapped to **1** (pure white).
- **Effect:**

- Very bright areas lose some detail, but in a mostly dark image, there are few such pixels.

The final image:

- Dark areas are more detailed.
- Overall contrast is improved.
- Very darkest and very brightest pixels are clipped.



$$s = C1 (I - C2)$$

where

- I = input intensity
- $C2$ = **shift constant** (brightness shift)
- $C1$ = **scaling constant** (contrast stretch)
- s = output intensity

Picture 1 - Histogram of I

- This is the **original image histogram**.
- Intensities lie in a small range (low contrast).
- No operation applied yet.

$$s = I$$

The image looks dull because intensity values are compressed.

Picture 2 - Histogram of $(I - 0.1)$

This corresponds to the **shift operation**.

From the formula: $I - C_2$

Here, $C_2 = 0.1$

What happens

- Each pixel intensity decreases by 0.1.
- Histogram shifts left.
- Shape does not change.

Purpose: move intensity range toward zero before stretching.

Picture 3 - Histogram of $4(I - 0.1)$

Now scaling is applied.

Full formula: $s = C_1 (I - C_2)$

Here, $C_1 = 4$, $C_2 = 0.1$

What happens

- After shifting, values are multiplied by 4.
- Histogram spreads across a wider range.
- Contrast increases.

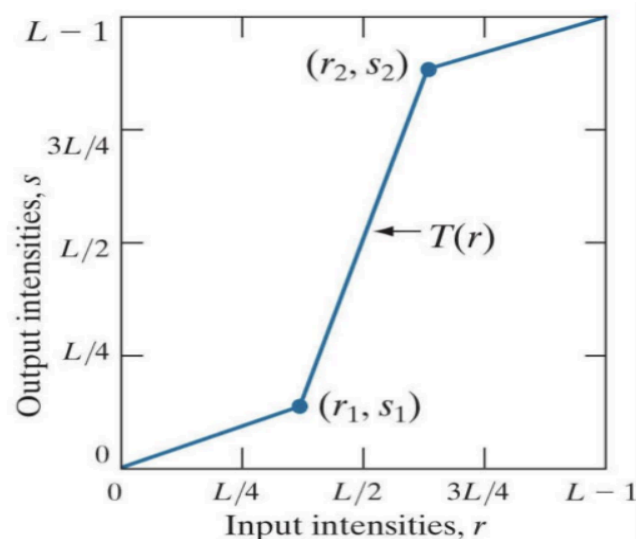
The image becomes clearer and more detailed.

What Shift and Scale Do

Constant	Operation	Effect on Histogram
C2	Shift (subtract)	Moves histogram left/right
C1	Scale (multiply)	Expands or compresses histogram

Contrast Stretching Using Piecewise Linear Mapping

This method improves image contrast by **stretching only the important intensity range** while compressing less important regions.



What the Graph Shows

- **X-axis (r)** $\rightarrow$ Input intensity
- **Y-axis (s)** $\rightarrow$ Output intensity
- $T(r) \rightarrow$ Transformation (mapping) function

The transformation curve is **piecewise linear**, meaning it consists of several straight-line segments.

Two control points define the mapping: (r_1, s_1) and (r_2, s_2)

These points divide the intensity range into **three regions**.

Three Mapping Regions

1. Dark Region: $([0, r_1] \rightarrow [0, s_1])$

Condition: $s_1 < r_1$

What happens

- Slope < 1
- Intensities are **compressed**.

Dark pixels change slightly.

Prevents over-enhancement of shadows and noise.

2. Middle Region: $([r_1, r_2] \rightarrow [s_1, s_2])$

This is the **most important region**.

What happens-

- Slope > 1
- Intensities are **stretched**.

Contrast increases significantly. Image details become clearer.

This region usually contains the most useful image information.

3. Bright Region: $([r_2, 1] \rightarrow [s_2, 1])$

Condition: $s_2 > r_2$

What happens

- Slope < 1 again
- Bright intensities are **compressed**.

Prevents highlight saturation.

Shape of the Transformation Curve

- Gentle slope $\rightarrow$ Compression
- Steep slope $\rightarrow$ Contrast stretching

Overall shape:

Flat $\rightarrow$ Steep $\rightarrow$ Flat

Meaning: Only the useful intensity range is enhanced.

No Information Loss : No information is removed because:

- Each input intensity (r) maps to **exactly one** output intensity (s).
- The function is:

- continuous
- monotonically increasing

$$r_1 < r_2 \rightarrow s_1 < s_2$$

So different input values remain different after transformation.

No Clipping Occurs

Clipping happens when intensity values exceed allowed limits and become identical, causing information loss.

In piecewise linear contrast stretching:

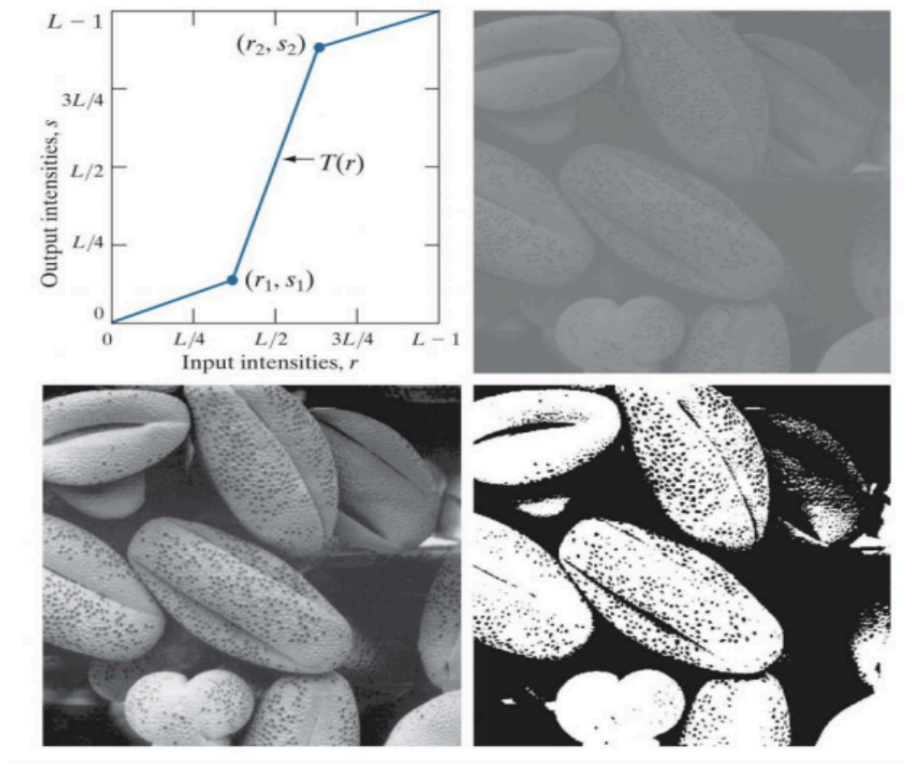
- Output values remain within $[0, L-1]$.
- No intensities are cut off.
- Values are only **redistributed**, not deleted.

Effect on Different Regions

Region	Operation	Result	Information Loss
Dark	Compression	Reduce noise	No
Middle	Stretching	Increase contrast	No
Bright	Compression	Avoid saturation	No

Compression $\neq$ deletion - spacing between intensity levels changes, but data remains.

Contrast Stretching Using Piecewise Linear Mapping



This picture shows **how piecewise linear contrast stretching works on a real image**, step by step - from transformation function $\rightarrow$ low-contrast image $\rightarrow$ enhanced image $\rightarrow$ thresholded image.

(a) Piecewise Linear Transformation Function

This graph defines the mapping:

- **x-axis (r)** $\rightarrow$ input intensity
- **y-axis (s)** $\rightarrow$ output intensity

- $T(r) \rightarrow$ transformation rule

Two points control the mapping: $(r_1, s_1), (r_2, s_2)$

What the curve does

- Dark intensities $\rightarrow$ compressed (small slope)
- Middle intensities $\rightarrow$ stretched (large slope)
- Bright intensities $\rightarrow$ compressed again

Only the important intensity range is expanded.

Purpose: increase contrast without losing information.

(b) Low-Contrast Electron Microscope Image

This is the **original image** of pollen.

Characteristics

- Gray and dull appearance
- Pixel intensities lie in a narrow range
- Details are hard to see

Histogram is compressed $\rightarrow$ **low contrast**.

(c) Result of Contrast Stretching

After applying the transformation $T(r)$:

What changes

- Middle intensity values are stretched.

- Differences between pixels increase.
- Texture and structure become clearer.

Observation

- Pollen surface details are now visible.
- The image looks sharper and more informative.
- Contrast improved

No clipping

No information loss

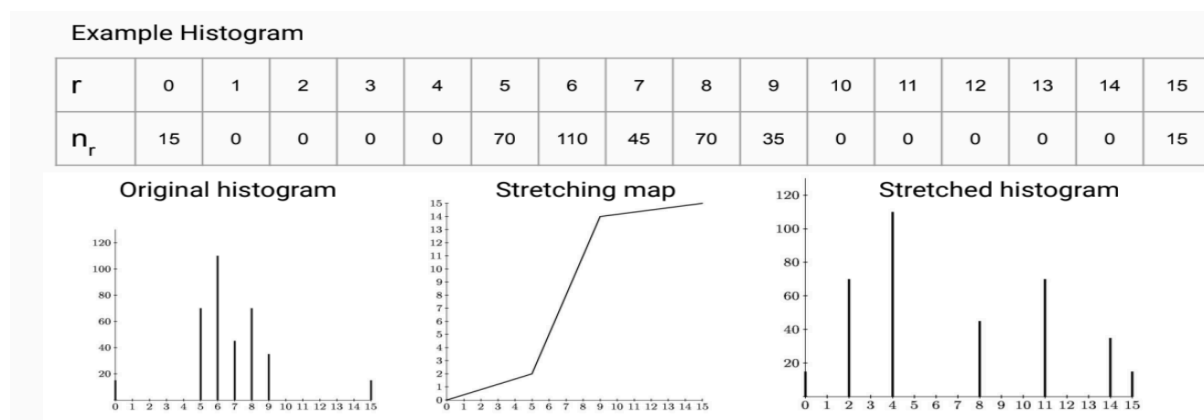
(d) Result of Thresholding

After contrast enhancement, a **threshold operation** is applied.

Result

- The image becomes **binary** (black & white).
- Objects are separated from the background.

Useful for segmentation and object detection.



1. Example Histogram

The table shows pixel intensity information.

- $(r) \rightarrow$ intensity levels (0-15)
- $(n_r) \rightarrow$ number of pixels at each intensity

Example:

- Intensity 6 $\rightarrow$ 110 pixels (highest)
- Intensities mainly between **5 and 9**
- Few pixels at 10 and 15

Meaning: most pixels are concentrated in a **small intensity range**. This indicates a **low-contrast image**.

2. Original Histogram

This graph plots the table values.

Observation

- Histogram is narrow.
- Pixels mostly lie between intensity **5-9**.
- Large parts of intensity range unused.

The image looks gray and dull because brightness values are crowded together.

3. Stretching Map

This is the **intensity transformation function**, $s = T(r)$.

- X-axis $\rightarrow$ input intensity (r)
- Y-axis $\rightarrow$ output intensity (s)

What the mapping does

♦ Low intensities (0–5)

- Small slope $\rightarrow$ compression

♦ Middle intensities (5–9)

- Steep slope $\rightarrow$ stretching
- Contrast increases here

♦ High intensities (9–15)

- Gentle slope $\rightarrow$ compression again

An important intensity region is expanded.

4. Stretched Histogram

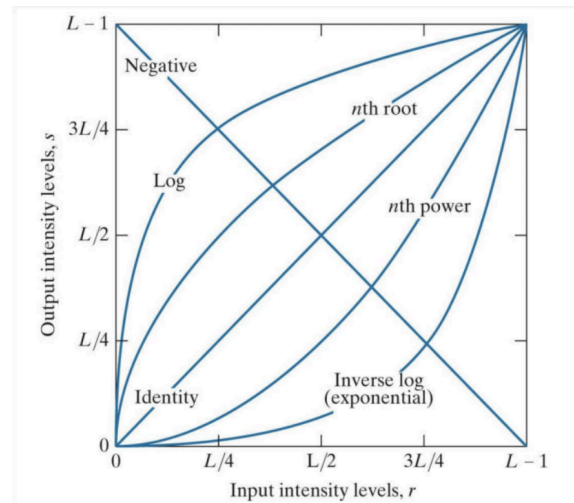
After applying the transformation:

What changes?

- Pixel values spread across a wider range.
- Histogram becomes wider.
- Contrast increases.

Now intensities occupy almost the full range **0–15**. Details become more visible.

Intensity Transformation - Basic Transformations



This picture shows **Basic Intensity Transformations** used in **Digital Image Processing**.

Graph Meaning

- **X-axis (r)** → Input intensity level
- **Y-axis (s)** → Output intensity level
- **(L-1)** → Maximum intensity value

Each curve represents a **different transformation function**: $s = T(r)$

Different shapes $\Rightarrow$ different brightness and contrast effects.

1. Identity Transformation

Straight diagonal line.

$$s = r$$

Meaning

- Output = Input
- The image remains unchanged.

No enhancement.

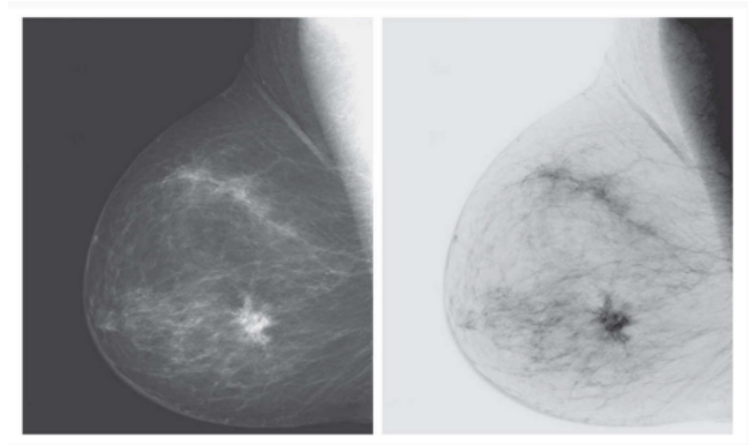
2. Negative Transformation

Line from top-left to bottom-right.

$$s = (L-1) - r$$

Effect

- Dark areas become bright.
- Bright areas become dark.



Useful for:

- medical images like x-ray
- highlighting white details on a dark background.

3. Log Transformation

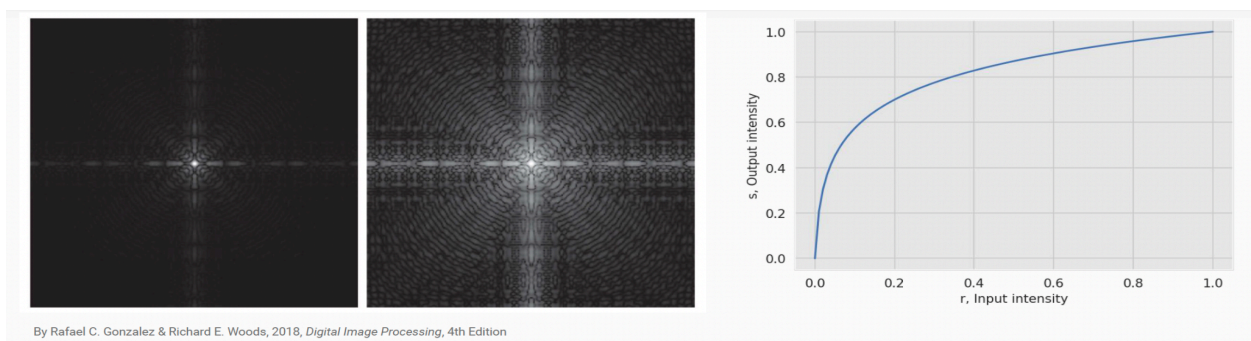


Fig: Visualization of fourier spectra of an image before and after log compression

The curve rises quickly at low intensities.

$$s = c \log(1+r); r \geq 0, c \geq 0$$

Effect

- Expands dark regions.
- Compresses bright regions.

Reveals details in dark areas.

Used when the image is mostly dark.

4. Inverse Log (Exponential)

Opposite of log curve.

Effect

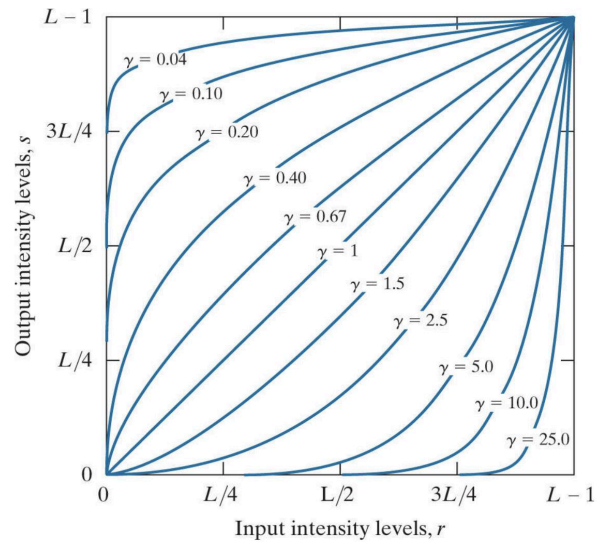
- Compresses dark values.
- Expands bright regions.

Enhances bright areas.

5. Power-Law Transformation (Gamma Transformation)

General form: $s = cr^\gamma$

- $\gamma < 1 \rightarrow$ expands dark intensities
- $\gamma > 1 \rightarrow$ expands bright intensities
- $\gamma = 1 \rightarrow$ identity (no change)



Two cases are shown:

(a) nth Root ($\gamma < 1$)

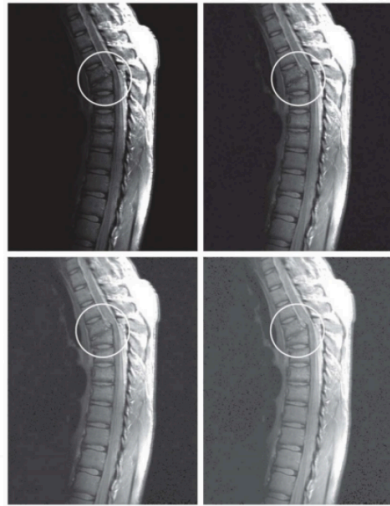
The transformation curve bends **upward**. Log transformation, dynamic range compression

Effect

- Brightens image.
- Low intensity values are expanded.
- Dark regions become brighter.
- Shadow details become more visible.

Used for:

- dark images
- display correction.



(a) Magnetic resonance image (MRI) of a fractured human spine (the region of the fracture is enclosed by the circle).

(b)–(d) Results of applying the transformation in Eq. (3-5) with $c = 1$ and $\gamma = 0.6, 0.4, 0.3$, respectively.

(Original image courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

In the original MRI:

- Most details lie in darker intensity regions.
- The fracture area is subtle.

When $\gamma < 1$:

- Small input intensities are mapped to larger output values.
- The lower dynamic range is expanded.
- Hidden dark details become visible.

That's why the fracture inside the circle becomes clearer.

Mathematically:

If $\gamma < 1$,

$$r^\gamma > r \text{ for } 0 < r < 1$$

So pixel values increase $\rightarrow$ image brightens.

(b) nth Power ($\gamma > 1$)

Curve bends downward. Inverse log transformation, dynamic range expansion.

Effect

- Darkens image.
- High intensity values are expanded.
- Low intensities are compressed.
- Bright regions get darker.

Used when the image is too bright.



(a) Ariel Image

(b)–(d) Results of applying the transformation in Eq. (3-5) with $c = 1$ and $\gamma = 3, 4, 5$, respectively.

(Original image courtesy of NAS)

Now the higher intensity range is emphasized.

If $\gamma > 1$,

$$r^\gamma < r \text{ for } 0 < r < 1$$

So:

- Mid-tones and shadows are compressed.
- Brighter areas are emphasized.

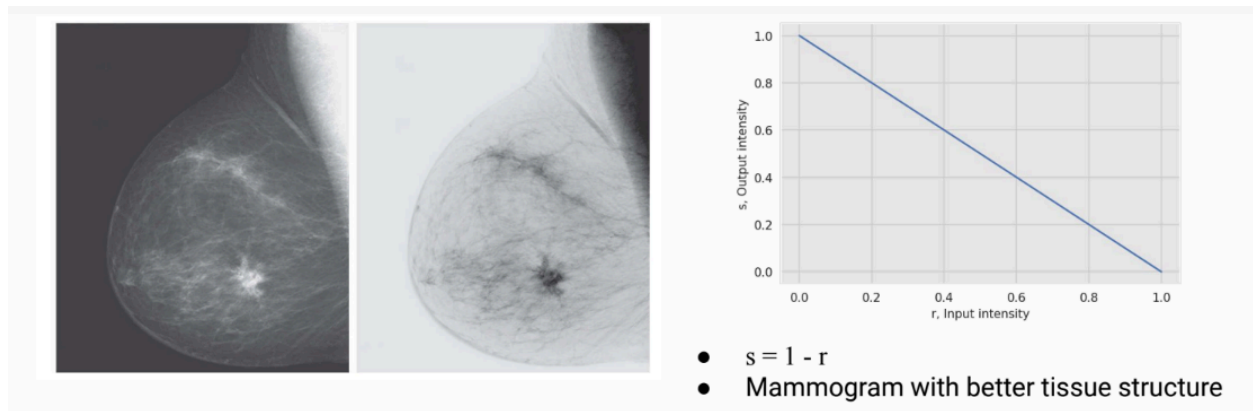
This improves contrast in bright structures (e.g., runway edges, buildings).

Key Idea of the picture

Different transformation curves redistribute intensity values differently:

- Steep near left → enhances dark pixels
- Steep near right → enhances bright pixels
- Straight line → no change

Intensity Transformation - Negative Image



This picture explains **Intensity Transformation using a Negative Image** - one of the simplest image processing operations.

Main Idea

A **negative transformation** reverses the brightness of an image:

- Bright pixels → become dark
- Dark pixels → become bright

Mathematically: $s = 1 - r$

or $(s = (L-1) - r)$ for (L) intensity levels

where:

- r = input intensity

- s = output intensity

Left Image (Original Image)

- This is the **original mammogram** image.
- Bright tissues appear white.
- Some structures are hard to distinguish from the background.

Contrast between tissues may not be clear.

Middle Image (Negative Image)

After applying the negative transformation:

- Dark regions become bright.
- Bright regions become dark.

Result

- Tissue structures become more visible.
- Fine details stand out better.

Helps doctors analyze mammograms more easily.

Right Graph (Transformation Function)

The graph shows how intensities change.

- X-axis $\rightarrow$ input intensity (r)
- Y-axis $\rightarrow$ output intensity (s)

The line slopes downward:

- $(r = 0 \Rightarrow s = 1)$ (black $\rightarrow$ white)
- $(r = 1 \Rightarrow s = 0)$ (white $\rightarrow$ black)

Complete intensity inversion.

Why Negative Image is Useful

Negative transformation is helpful when:

- Important details are in **dark regions**
- Background is bright
- Medical imaging (X-ray, mammogram)
- Satellite images

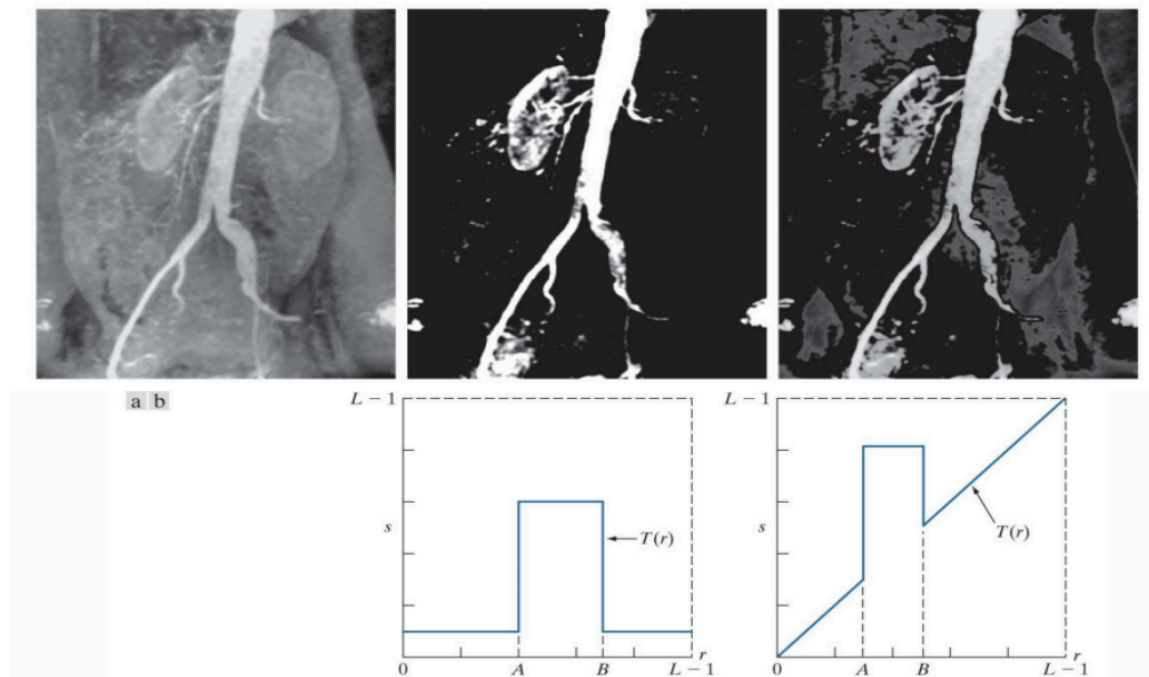
It improves visibility without changing image structure.

What Actually Happens

Original	After Negative
Dark pixel	Bright pixel
Bright pixel	Dark pixel
Structure same	Only brightness inverted

No information is lost - only intensity values are reversed.

Gray-Level Slicing



Gray-level slicing is an image enhancement technique used to **highlight a specific range of intensity values** in an image.

Instead of modifying all pixels, we focus only on pixels whose gray levels fall within a chosen interval $[A, B]$.

It is commonly used in:

- Medical imaging (highlight vessels, tumors)
- Remote sensing
- Industrial inspection

Analyze the Images

(Left Image) – Original Medical Image

This looks like an angiographic image where blood vessels are visible but blended with surrounding tissues. The vessels have intensity values within a specific gray-level range.

(Middle Image) – Gray-Level Slicing (Background Suppressed)

Here:

- Pixels with intensity between **A and B** are set to a high value (bright).
- All other pixels are set to low (dark).

Result:

- Only vessels remain visible.
- Background tissues disappear.

This is **binary-style slicing** (hard highlighting).

(Right Image) – Gray-Level Slicing (Background Preserved)

Here:

- Pixels in range $[A, B]$ are enhanced (brightened).
- Other pixels remain unchanged.

Result:

- Vessels are emphasized.
- The background structure is still visible.

This is softer and more practical for diagnosis.

The Plots

The bottom graphs show the transformation function, $s = T(r)$.

Where:

- r = input gray level
- s = output gray level
- $L-1$ = maximum gray level
- A, B = selected intensity range

Left Plot (Without Background)

This is a **rectangular transformation**.

Behavior:

$$T(r) = \begin{cases} \text{high value} & A \leq r \leq B \\ \text{low value} & \text{otherwise} \end{cases}$$

Interpretation:

- Intensities between A and $B \rightarrow$ mapped to bright.
- Everything else $\rightarrow$ mapped to dark.

This produces the middle image (strong isolation).

Right Plot (With Background Preserved)

This transformation combines:

- Identity mapping outside the slice.
- Boosting inside the slice.

Behavior:

- For $r < A$ and $r > B$: $s = r$ (unchanged)
- For $A \leq r \leq B$: boosted to higher intensity

Graphically:

- Linear increase until A .
- Jump up in region $A-B$.
- Then continues linear again.

This produces the rightmost image.

What Is Actually Happening?

Gray-level slicing is a form of **selective dynamic range expansion**. Instead of expanding dark or bright regions globally (like gamma correction), it:

- Expands contrast only in a chosen interval
- Compresses or suppresses other regions

Why It Works Well for Medical Images

In angiography:

- Blood vessels often occupy a narrow intensity band.
- Surrounding tissue has different intensities.

So selecting $[A, B]$ isolates vessels effectively.

Key Difference from Thresholding

- Thresholding $\rightarrow$ divides image into binary regions.
- Gray-level slicing $\rightarrow$ emphasizes a range, not just above/below a threshold.